

Project Brief: The "CartShare" Challenge

Mission: Building a Collaborative Shopping Ecosystem

1. The Real-World Problem (Why are we doing this?)

In shared living or working environments—like student dorms, office teams, or travel groups—coordinating joint purchases is often chaotic.

- **The Communication Gap:** Groups currently use fragmented tools (WhatsApp, sticky notes, or spreadsheets) to track who needs what. This leads to missed items, duplicate purchases, and constant backtracking.
- **The Delivery Threshold Trap:** Online delivery platforms often require a minimum total cost (e.g., \$75) to unlock free shipping. Coordinating these orders manually is frustrating and prone to errors.
- **Lack of Transparency:** It is difficult to track who added which item, making it tedious to split expenses fairly when the order arrives.

2. Your Mission (The Goal)

Your task is to take the provided frontend prototype and evolve it into a **collaborative, real-time web application**. Students should be able to:

- Create or join a unique "Room" using a code.
- Add/remove items from a shared cart that updates for everyone in the room.
- View a real-time activity log of what participants are doing.
- Generate an audit-ready, printable receipt for the group.

3. Technical Requirements (Your Toolkit)

You will apply the skills from your syllabus to bridge the gap between static design and active functionality:

Requirement	Implementation
User Access	Use HTML forms and JavaScript to handle "login" and Room joining.
Responsive UI	Use CSS Flexbox, Grid, and Bootstrap to ensure the cart works on both phones and PCs.
Data Persistence	Use the browser's storage to keep cart data alive as items are added.
Collaboration	Use JavaScript event listeners to sync data across browser tabs.

Requirement	Implementation
Printing	Use CSS print media rules and JavaScript to generate a clean receipt summary.

4. Success Criteria

Your project will be evaluated on:

- **Functionality:** Does the cart update correctly when items are added?
- **Design:** Is the application fully responsive (looks good on mobile and desktop)?
- **Collaboration:** Can you successfully simulate multiple users in the same room?
- **Submission:** Are your files organized according to the folder structure guidelines in our course?

5.Submission Guidelines

1.Repository Setup: Host your project on GitHub and provide the repository link. Ensure you have a README.md file that explains how to run the application locally and documents the features you implemented.

2.Live Deployment: Deploy your application using Vercel, Netlify, or GitHub Pages. Submit the URL to your live project so that the "Room" functionality can be tested across multiple browser sessions.

3.Naming Convention: When submitting your project link via the portal, use the format: BatchID_FullName_CartShare. Ensure your folder structure strictly adheres to the css/, js/, and assets/ organization outlined in the course syllabus.